Sahaj Bhadja

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Summary

MSCS graduate (May 2026) passionate about **open source** and **Linux/Ubuntu**, with strong academics and hands-on work in **Python**, **C/C++**, **Go**, **Bash**, containers, and performance-focused systems software on GitHub.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Software Engineering, Analysis of Algorithms, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built resilient **Python services** and data pipelines on **Linux/cloud** infrastructure for 500+ users, optimizing performance from **1.2s to 350ms** through profiling, SQL tuning, and secure production operations.
- Maintained CI/CD releases and automated validation for 5K+ monthly jobs, engaging with issue-driven debugging and documentation to improve reliability and code quality in a distributed team.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Delivered **Python backend** features and APIs serving 10K+ daily requests, participating in code reviews, user-focused fixes, and maintainable open-style engineering practices in a fast-moving product team.
- Improved system responsiveness by **60%** through performance analysis and production troubleshooting across Linux-hosted deployment environments.

USC AD Labs **Los Angeles, CA**
Software Engineering Assistant *January 2025 – May 2025*

- Implemented **Python data pipelines** with reproducible test harnesses across 100K+ records, emphasizing correctness, observability, and iterative improvement in research engineering workflows.
- Authored clear technical notes and review feedback for collaborators, demonstrating accountability, curiosity, and reflective engineering communication.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform software and CI/CD pipelines across **Linux, Windows, and macOS**, improving build/test automation reliability by **68%** with Jenkins, PyTest, and Git-based workflows.
- Developed **Python microservices** and API integrations with security-aware design, reducing response latency by **30%** and supporting enterprise-grade data movement on Linux servers.

OpenStack Labs **India**
Open Source Engineering Intern *January 2023 – April 2023*

- Contributed to **Ubuntu/Linux** tooling with **Bash and Python** scripts for packaging workflows, environment setup, and automated validation in open source-style collaboration.
- Participated in issue tracker discussions, code reviews, and documentation updates while learning distribution concepts and developer-facing command-line tooling.

Skills

Languages: **Python**, **C/C++**, **Go**, JavaScript, Java, **Bash**, SQL
Linux & OSS: **Ubuntu/Linux**, Docker, containers, Git/GitHub, CLI tooling, packaging concepts, issue-driven development
Systems: Operating systems fundamentals, performance tuning, security-aware coding, CI/CD (Jenkins), networking basics

Projects

CPU Scheduling Simulator Live | *C++, Systems Programming*

- Built a **low-level scheduling simulator** (FCFS, SJF, Round Robin, MLQ) demonstrating kernel-adjacent resource management, performance trade-offs, and systems design fundamentals.

Chat2SQL Code | *Java, Linux, Docker, Open Source*

- Maintained an **open source** cross-platform developer tool supporting 10+ databases with Linux/macOS/Windows workflows, Maven builds, and community-style code quality practices.

Linguistix Code | *Python, Docker, Linux Deployment*

- Published a **containerized Python service** with automated validation and Linux-friendly deployment patterns, emphasizing resilient APIs and operational observability.